Spinal Cord Anatomy & Clinical Correlates

Post-Class Slides

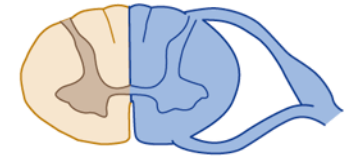
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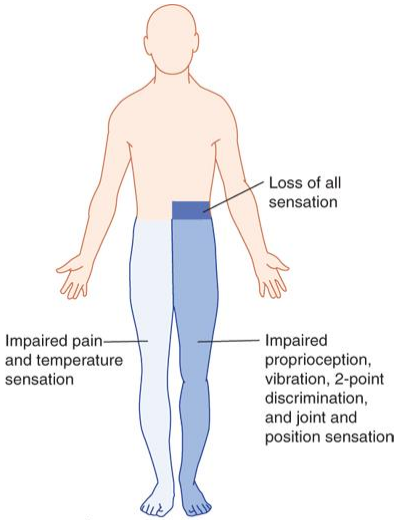
Spinal Cord

Clinical Correlates

Brown Séquard Syndrome (Hemisection)

Cause: Spinal cord hemisection; classically penetrating trauma



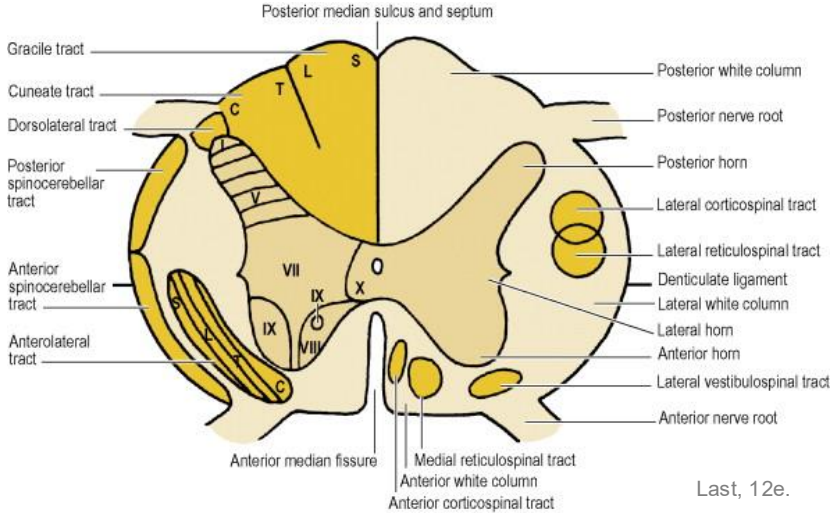


At the Lesion

- Ipsilateral LMN signs (anterior horn)
- Ipsilateral complete sensory loss at lesion level

Below the Lesion

- Ipsilateral UMN signs (corticospinal tract)
- Ipsilateral loss of fine touch, vibration, proprioception (dorsal column)
- Contralateral loss of pain, temperature (spinothalamic tract)



CST + DCML = ipsilateral | Spinothalamic = contralateral

Anterior Spinal Artery (ASA) Syndrome

Cause

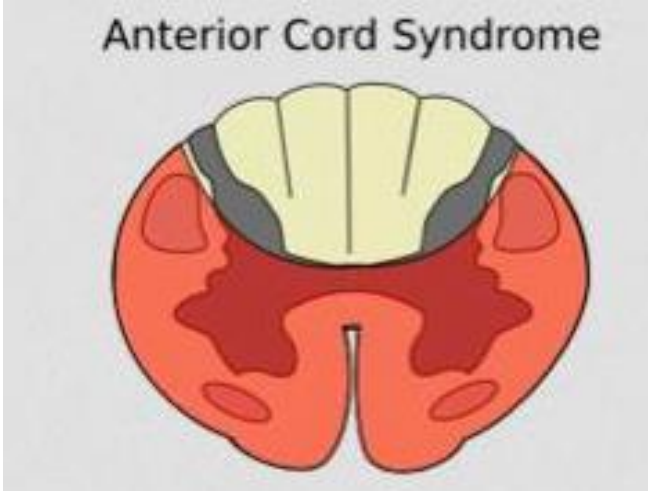
- Infarction of anterior 2/3 of spinal cord
- May occur with aortic pathology or surgery, ASA occlusion

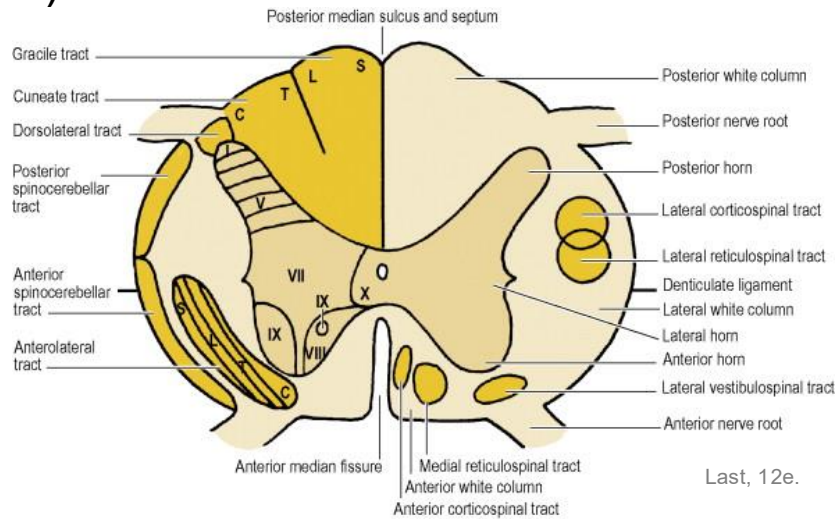
Below the Lesion

- Bilateral UMN weakness (corticospinal tracts)
- Bilateral loss of pain, temperature (spinothalamic tracts)
- Vibration, proprioception, fine touch preserved (dorsal column spared)
- ± Autonomic dysfunction (bladder, bowel, sexual, hypotension)

At the Lesion Level

- LMN weakness (anterior horn)

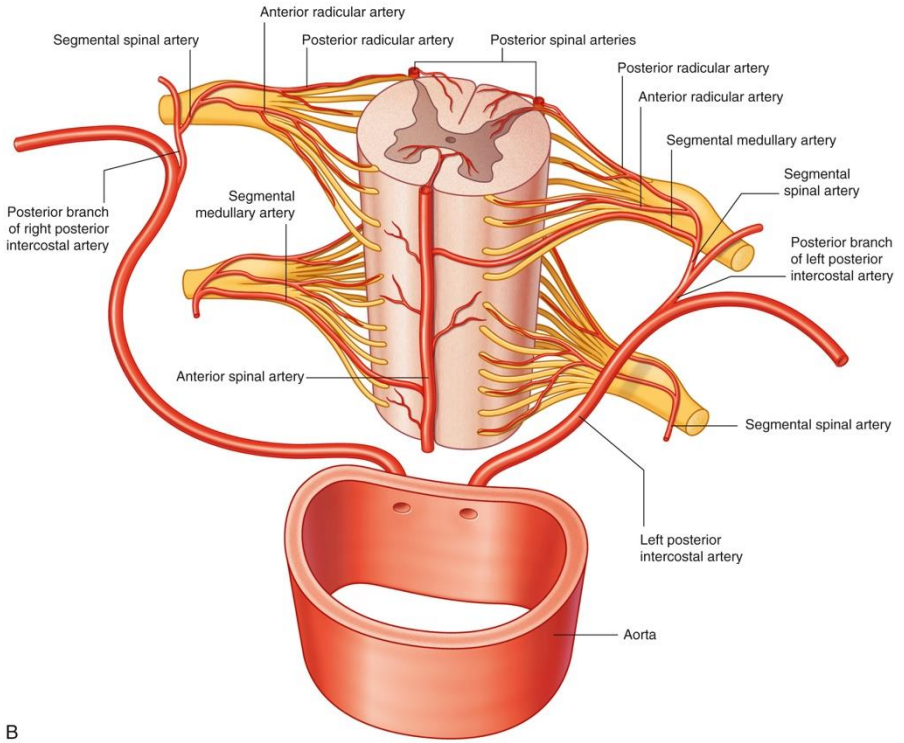


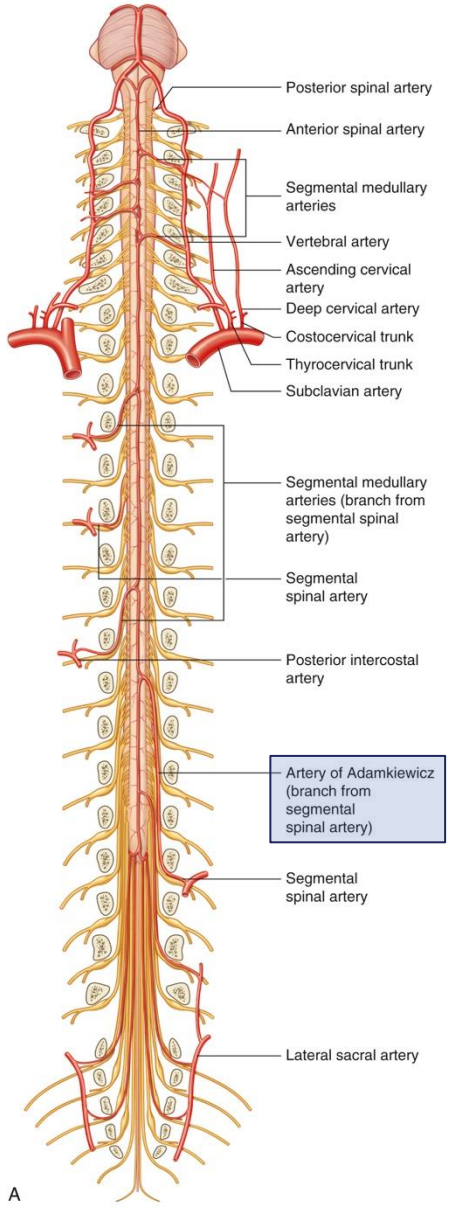


Last, 12e.

Artery of Adamkiewicz

- Major segmental medullary artery
- Reinforces blood supply to the lower spinal cord
- Usually arises from the lower thoracic/upper lumbar region (T9–L2)
- Injury during aortic surgery may cause anterior spinal artery syndrome

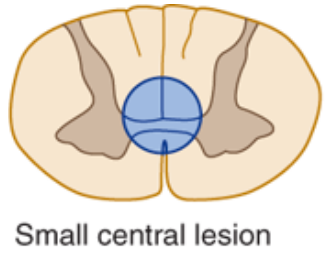


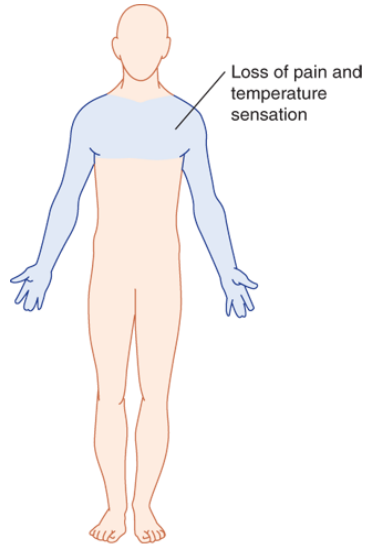


Arterial Supply of Spinal Cord
Fig. 2.55. GAS, 5e.

Syringomyelia

- Fluid-filled cavity (syrinx) within the central spinal cord
- Most commonly affects cervical cord
- Damages crossing spinothalamic fibers in the anterior white commissure
- → Bilateral pain/temperature loss in a cape-like distribution
- Vibration/proprioception preserved



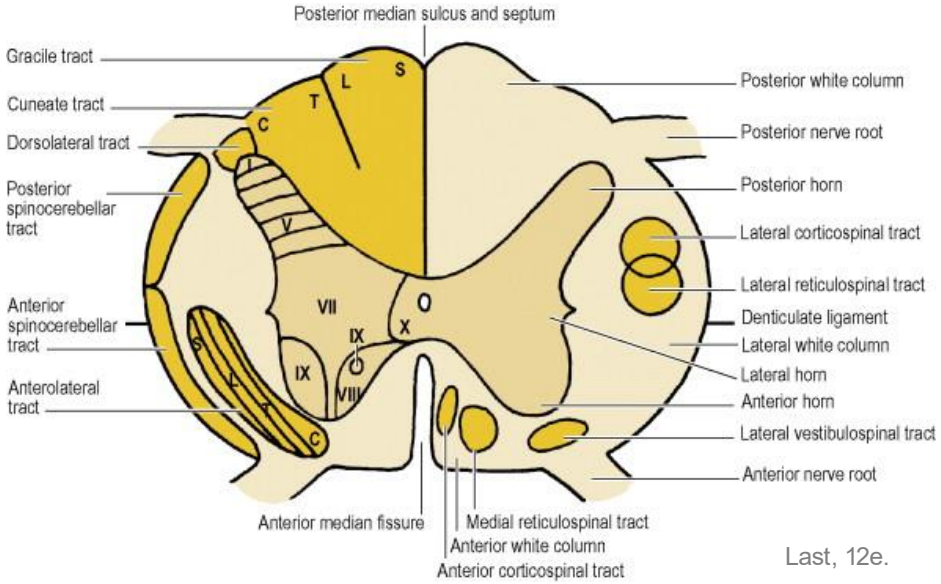


With expansion

- Anterior horn → LMN weakness/hand muscle atrophy
- Lateral horn/sympathetic pathway → Horner syndrome
- Corticospinal tract → UMN signs below lesion

Association

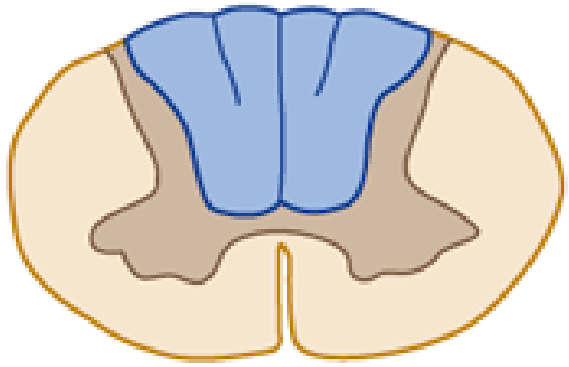
- Chiari I malformation



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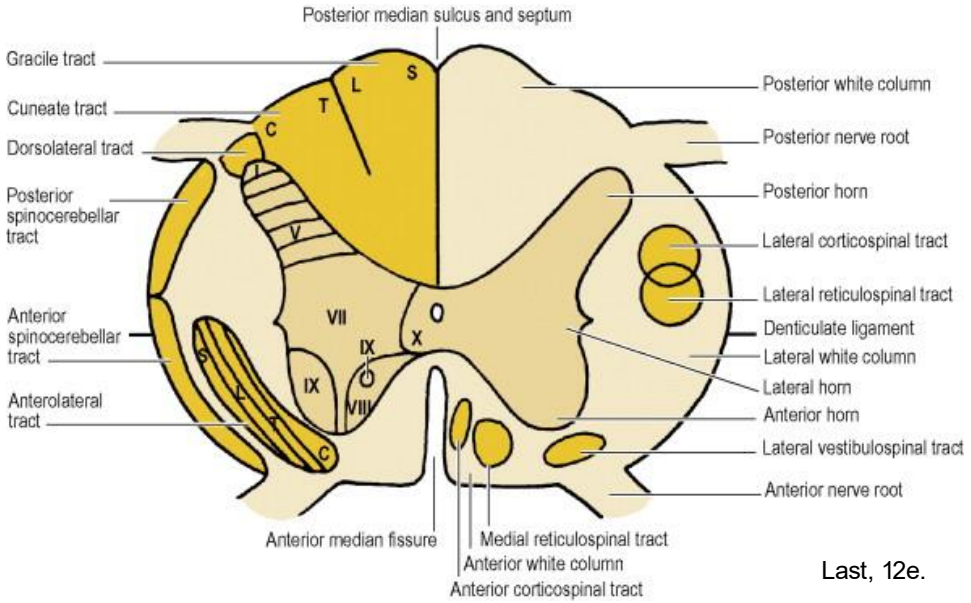
Subacute Combined Degeneration

- Associated with vitamin B12 deficiency
- Demyelination of:
  - Dorsal columns
  - Lateral corticospinal tracts
- ↓ Vibration, proprioception → sensory ataxia, positive Romberg
- Corticospinal involvement → spastic weakness, hyperreflexia, Babinski
- May cause paresthesias and peripheral neuropathy



Dorsal column lesion

B12 deficiency → dorsal columns + corticospinal tracts

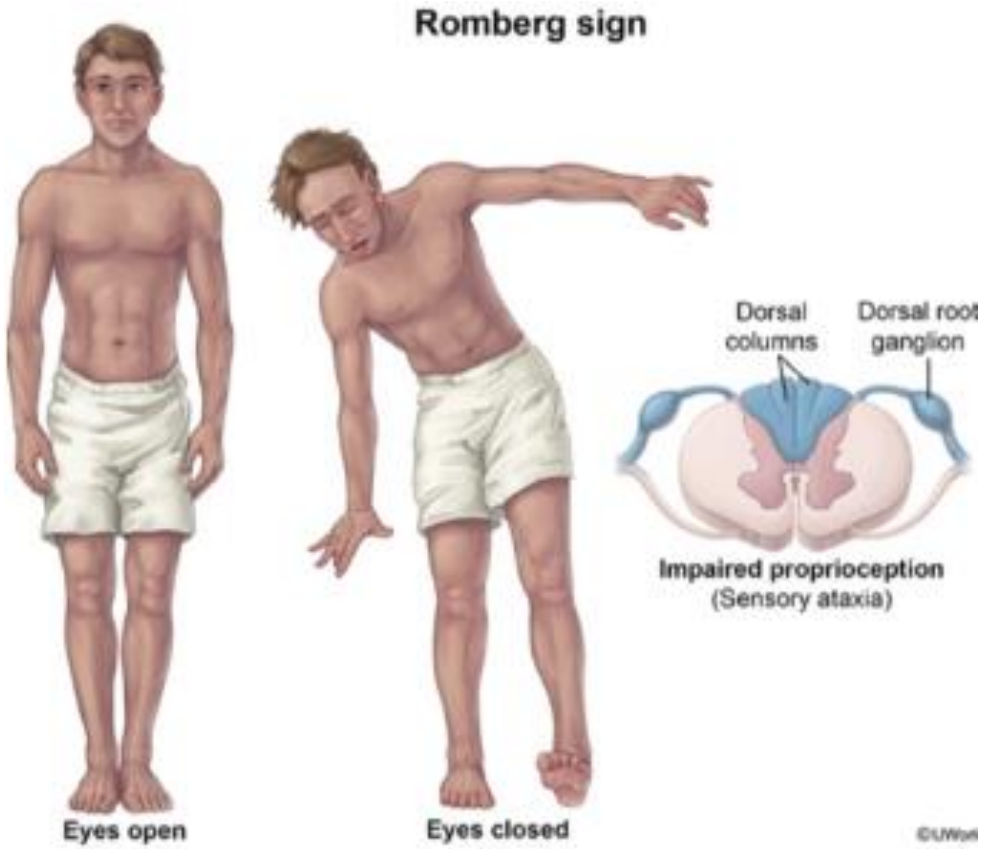


Last, 12e.

Ataxia

Ataxia = impaired coordination, not weakness

- **Sensory:** impaired proprioceptive → worse with eyes closed; positive Romberg
- **Cerebellar:** wide-based gait, dysmetria, intention tremor
- **Vestibular:** vertigo, nystagmus, imbalance

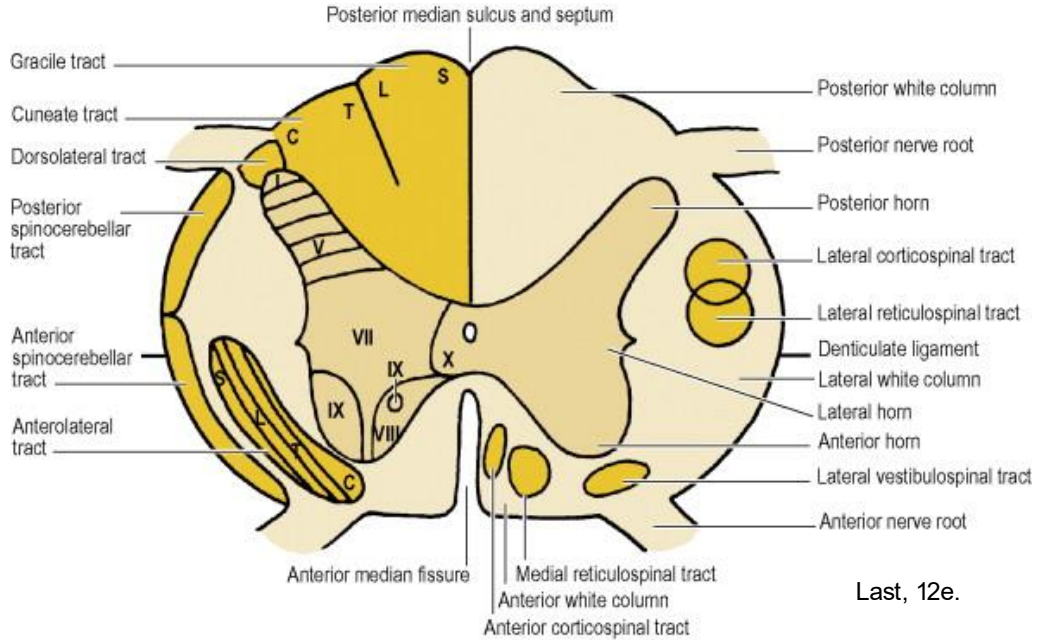


Tabes Dorsalis

- Late manifestation of neurosyphilis
- Degeneration of dorsal columns + dorsal roots
- ↓ Vibration/proprioception → sensory ataxia, positive Romberg
- Dorsal root involvement → absent reflexes + lightning pains
- May cause urinary dysfunction (loss of bladder fullness sensation → dorsal root involvement)
- Argyll Robertson pupils: accommodate but do not react to light

SCD → dorsal columns + **corticospinal tracts**

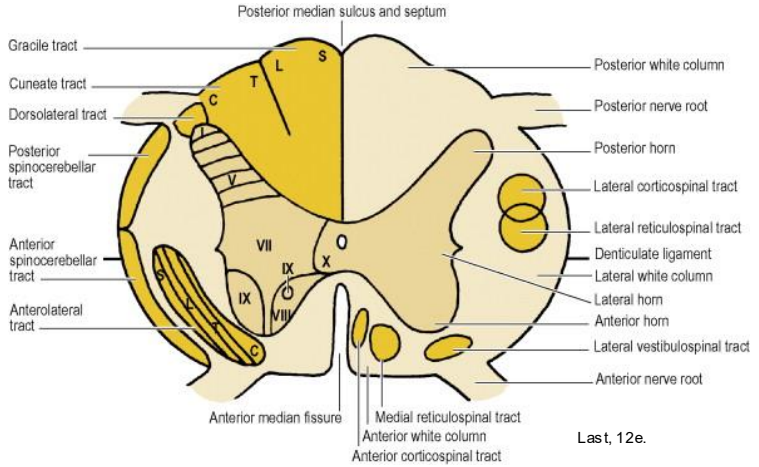
Tabes dorsalis → dorsal columns + **dorsal roots**



Last, 12e.

Friedreich Ataxia

- Autosomal recessive; GAA repeat expansion in frataxin



Last, 12e.

Degeneration of:

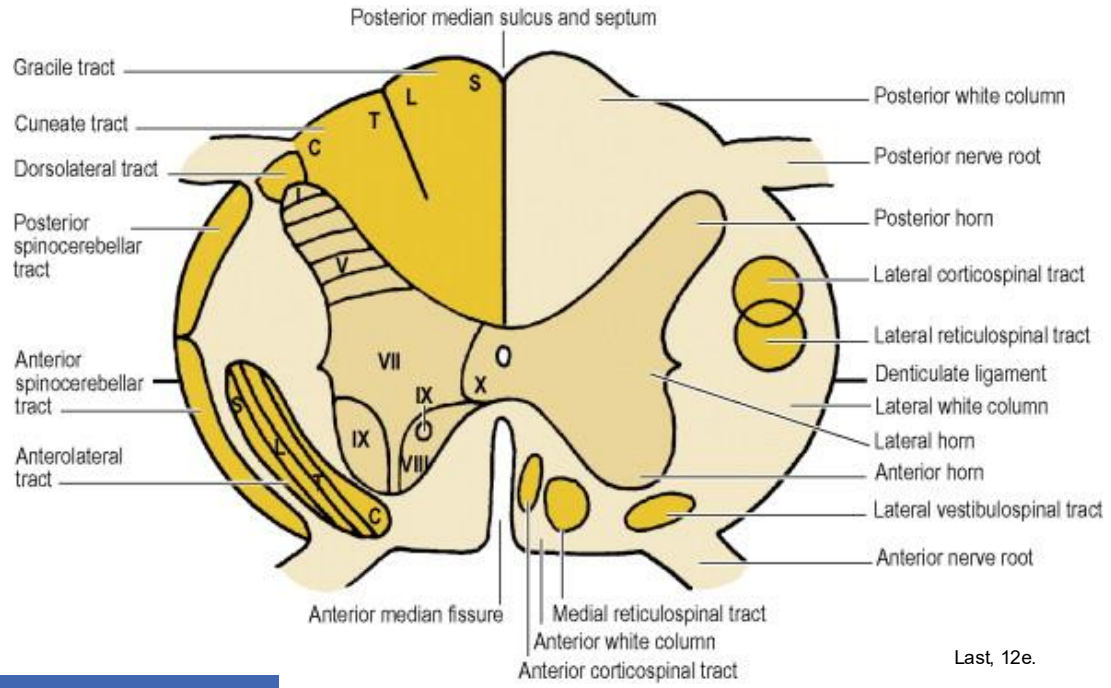
- Dorsal columns → ↓ vibration/proprioception → sensory ataxia
- Spinocerebellar tracts → ↓ proprioceptive input to cerebellum → gait/limb ataxia
- Corticospinal tracts → UMN signs (spasticity, Babinski)

- Absent DTRs → impaired sensory afferent limb of reflex arc
- Associated with hypertrophic cardiomyopathy and diabetes
- Onset in childhood/adolescence

SCD → dorsal columns + **corticospinal tracts**
Tabes dorsalis → dorsal columns + **dorsal roots**
Friedreich → dorsal columns + **spinocerebellar + corticospinal tracts**

Poliomyelitis

- Poliovirus selectively damages anterior horn motor neurons
- Produces LMN signs
  - Flaccid weakness/paralysis
  - ↓ Muscle tone
  - ↓ Reflexes
  - Muscle atrophy
- Sensation is preserved
- Weakness may be asymmetric



Anterior horn lesion → LMN signs with preserved sensation

Spinal Muscular Atrophy (SMA)

- Autosomal recessive disorder due to SMN1 (Survival Motor Neuron 1) gene deletion/mutation
- Degeneration of anterior horn motor neurons
- Produces LMN signs: weakness, hypotonia, muscle atrophy, ↓ reflexes
- Sensation preserved

SMA Type I (Werdnig-Hoffmann Disease)

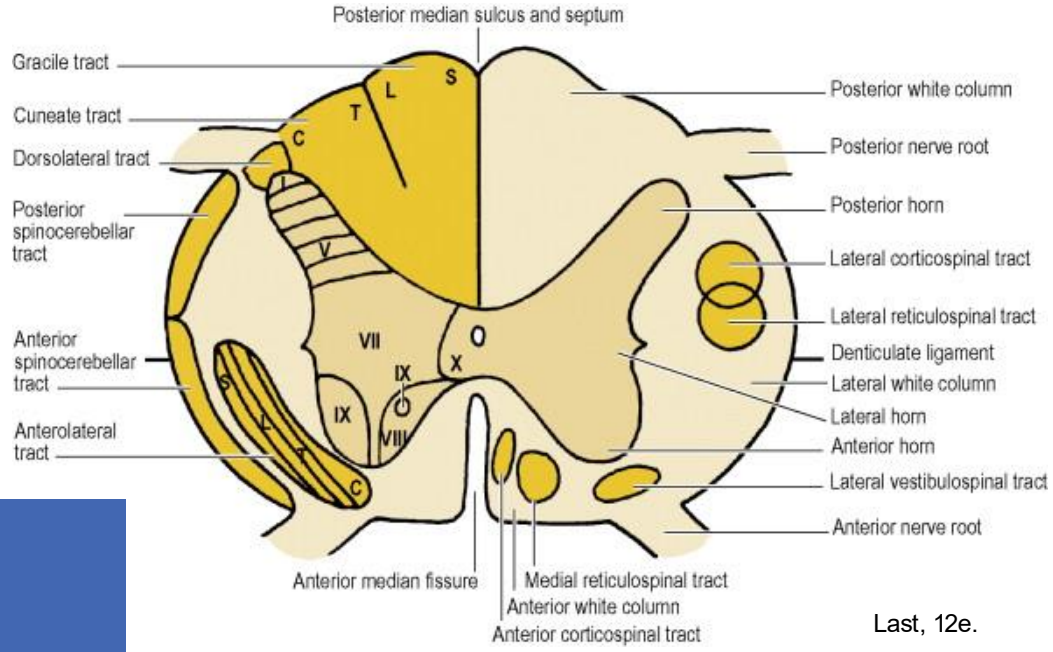
- Severe infantile form
- "Floppy infant"
- Symmetric proximal weakness
- Tongue fasciculations
- Respiratory weakness may occur



Poliomyelitis: viral destruction of anterior horn cells
SMA: genetic degeneration of anterior horn cells
Both → LMN signs with preserved sensation

Amyotrophic Lateral Sclerosis (ALS) / Lou Gehrig's Disease

- Progressive degeneration of:
  - Lateral corticospinal tracts (UMN) → spasticity, hyperreflexia, Babinski
  - Anterior horn motor neurons (LMN) → weakness, atrophy, fasciculations
- Mixed UMN + LMN signs
- Often begins asymmetrically
- **Sensation is preserved**
- Bulbar involvement → dysarthria, dysphagia
- Respiratory muscle weakness may occur

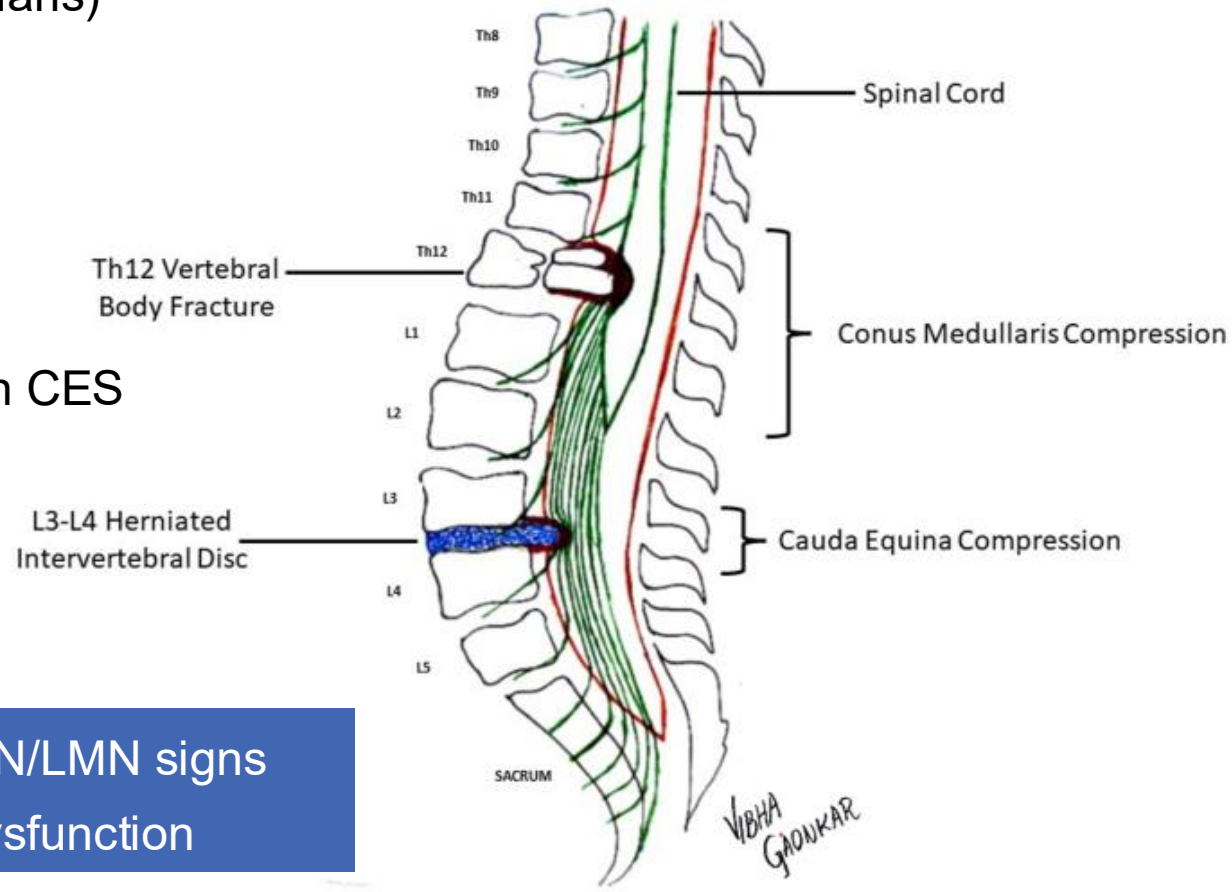


Last, 12e.

Polio → anterior horn → LMN
SMA → anterior horn → LMN
ALS → anterior horn + corticospinal tract → LMN + UMN

Conus Medullaris Syndrome (CMS)

- Lesion of the terminal spinal cord (conus medullaris)
- Findings usually bilateral and symmetric
- May produce mixed UMN + LMN signs
- Saddle anesthesia
- Early bladder/bowel dysfunction
- Radicular pain is typically less prominent than in CES



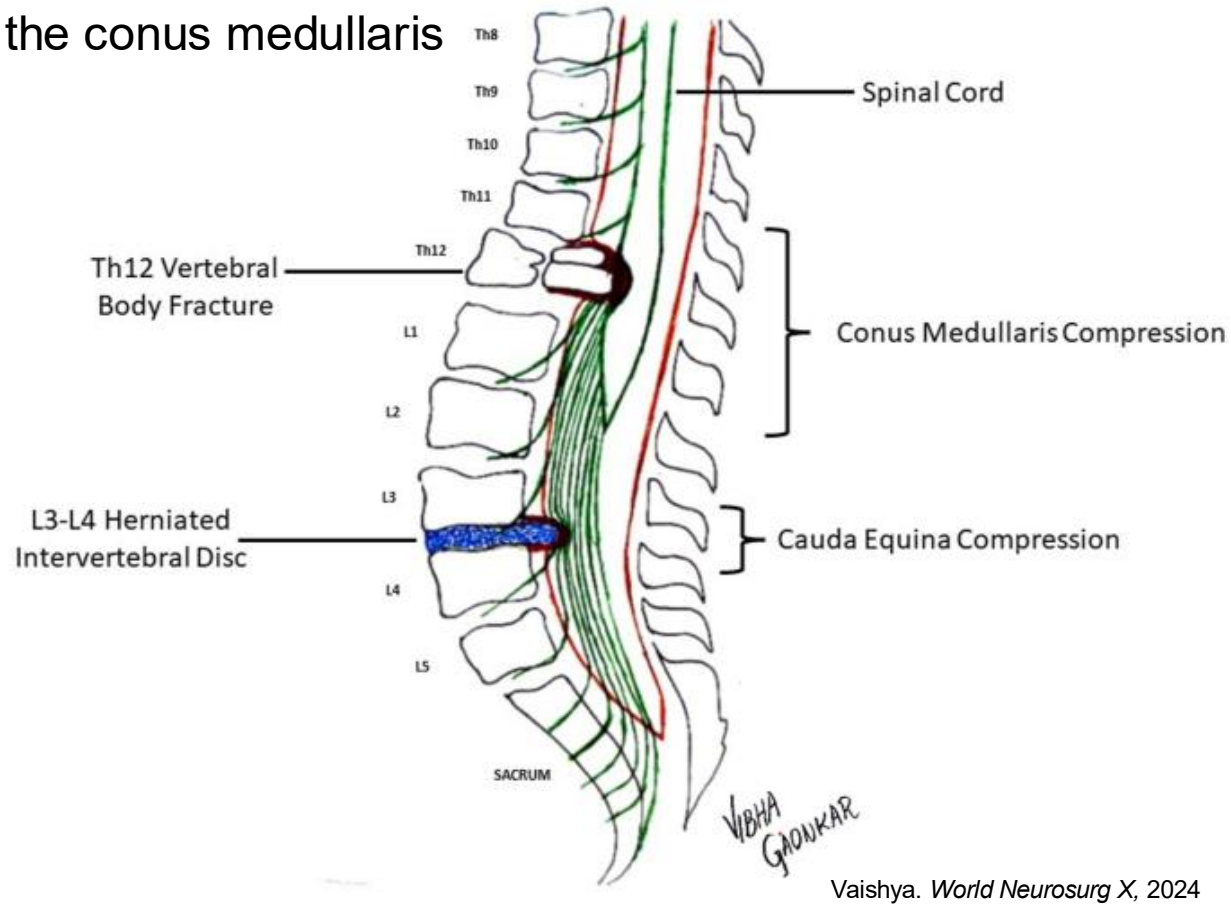
Vaishya. *World Neurosurg X,* 2024

Conus medullaris = spinal cord → may have UMN/LMN signs

Typically symmetric findings + early sphincter dysfunction

Cauda Equina Syndrome (CES)

- Compression of lumbosacral nerve roots below the conus medullaris
- LMN signs
  - Flaccid weakness
  - ↓/absent reflexes
- Sensory loss often asymmetric
- Saddle anesthesia
- Radicular low back/leg pain
- Bladder/bowel dysfunction may occur



Vaishya. *World Neurosurg X,* 2024

Cauda equina = nerve roots
→ asymmetric LMN findings

Conus = cord → may have UMN + LMN signs; symmetric; early sphincter dysfunction
Cauda equina = roots → LMN only; asymmetric; prominent radicular pain

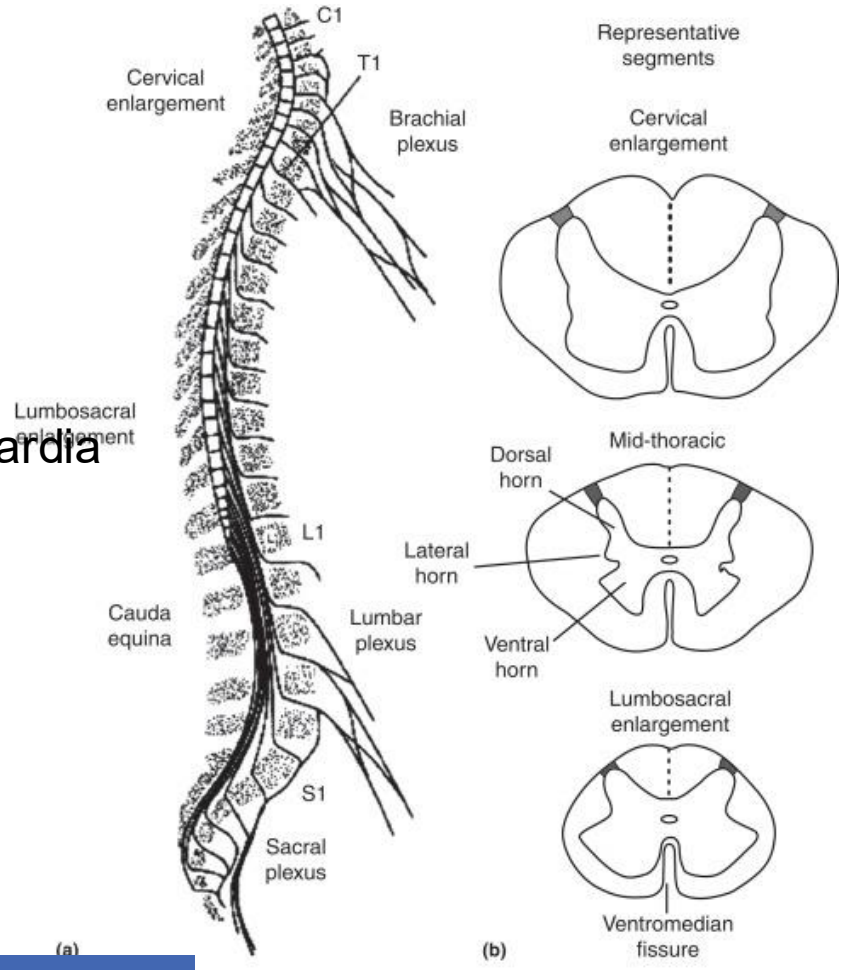
Complete Spinal Cord Transection

Acute Phase — Spinal Shock

- Loss of motor and sensory function below the lesion
- Flaccid paralysis
- Areflexia
- Loss of autonomic function
- Cervical/high thoracic injury may cause hypotension + bradycardia

After Spinal Shock Resolves

- Spasticity
- Hyperreflexia
- Babinski signs
- Persistent paralysis below the lesion



Acute cord injury → spinal shock: flaccid + areflexic
Later → UMN signs: spastic + hyperreflexic + Babinski

Summary

- Localize: modality + side + level
- UMN: spasticity, hyperreflexia, Babinski
- LMN: flaccidity, hyporeflexia, atrophy/fasciculations
- Tract location + decussation → pattern of deficits
- Selective tract disease → characteristic combinations of deficits
- Cord lesions → UMN ± LMN | Root lesions → LMN
- Spinal shock: acute areflexia/flaccidity → later UMN signs

Spinal Cord Anatomy & Clinical Correlates

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